NEWTON SCHOOL

PRE-CALCULUS | TEST PREP

Test Prep — Induction · Recursion · Series · Bernoulli

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Standard: *CCSS.MATH.CONTENT.HSF-IF.A.3 (sequences/recursion) · Mathematical Induction (PCBCH "Mr. Jones" 12-problem handout, 2026-05-07)*

KENNY'S REMINDER — WHAT THIS LECTURE COVERS

Tonight is a **review for the test**, built from the PCBCH (Mr. Jones) 12-problem induction handout (5/7) and our 5/13 board notes. Four topics: *Induction* · *Run-of-Heads recursion* · *Series (Σ notation, incl. Domain of Convergence)* · *Bernoulli's Madness* (power sums · hockey stick · telescoping · linear recurrence). For each topic: read the concept, work the example, then try each practice problem first — the answer sits right below it, so don't peek.

Topic 1 — Mathematical Induction (handout level)

The 3-step framework

- **Base case:** verify at the smallest valid n (usually n = 1, but sometimes n = 2 or n = 4 — when an inequality only starts working past a threshold).
- **Inductive hypothesis:** assume the statement holds for n = k.
- **Inductive step:** show it then holds for n = k + 1, using the hypothesis as your lever.

Patterns you must own from the Mr. Jones handout:

- Closed-form sum identities — e.g. $\sum k^3 = n^2(n+1)^2/4$.
- Divisibility — e.g. $13^n - 5^n$ is a multiple of 8 (the trick: split the larger base as smaller + multiple).
- Inequalities with a starting value — e.g. $n! > n^2$ from n = 4 onward.
- Fibonacci identities (#10–12) — these need *strong induction*: assume true for all n ≤ k, prove for k + 1.

Worked Example — $\Sigma k^3 = n^2(n+1)^2/4$ (PCBCH #2)

Base (n = 1): LHS $= 1^3 = 1$; RHS $= \dfrac{1^2 \cdot 2^2}{4} = 1$. ✓

Hypothesis (n = k): assume $1^3 + 2^3 + \cdots + k^3 = \dfrac{k^2(k+1)^2}{4}$.

Step (n = k + 1): add $(k+1)^3$ to both sides:

$$\frac{k^2(k+1)^2}{4} + (k+1)^3 = (k+1)^2\left[\frac{k^2}{4} + (k+1)\right] = \frac{(k+1)^2(k^2+4k+4)}{4} = \frac{(k+1)^2(k+2)^2}{4}. \checkmark$$

Practice — Topic 1

1(a) Prove for every positive integer n: $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$. *— PCBCH #1*

ANSWER — INDUCTION

Base (n = 1): $1 \cdot 1! = 1$ and $2! - 1 = 1$. ✓

Hypothesis (n = k): $1 \cdot 1! + \cdots + k \cdot k! = (k+1)! - 1$.

Step: add $(k+1)(k+1)!$ to both sides:

$$\left[(k+1)! - 1\right] + (k+1)(k+1)! = (k+1)!\left[1 + (k+1)\right] - 1 = (k+1)!(k+2) - 1 = (k+2)! - 1. \checkmark$$

1(b) Prove for every positive integer n: $13^n - 5^n$ is a multiple of 8. *— PCBCH #3*

ANSWER — DIVISIBILITY (SPLIT 13 = 5 + 8)

Base (n = 1): $13 - 5 = 8 = 8 \cdot 1$. ✓

Hypothesis (n = k): $13^k - 5^k = 8m$ for some integer m.

Step:

$$13^{k+1} - 5^{k+1} = 13 \cdot 13^k - 5 \cdot 5^k = 13(13^k - 5^k) + 5^k(13 - 5) = 13(8m) + 8 \cdot 5^k = 8\left(13m + 5^k\right). \checkmark$$

1(c) Find the obvious starting value, then prove: $n! > n^2$ from that point on. *— PCBCH #5*

ANSWER — INEQUALITY WITH STARTING VALUE

Try small n: $3! = 6 < 9 = 3^2$ (fails), but $4! = 24 > 16 = 4^2$ (works). Starting value $\mathbf{n = 4}$.

Hypothesis (n = k, k ≥ 4): $k! > k^2$.

Step:

$$(k+1)! = (k+1) \cdot k! > (k+1) \cdot k^2.$$

It is enough to show $(k+1)k^2 \geq (k+1)^2$, i.e. $k^2 \geq k+1$, i.e. $k^2 - k - 1 \geq 0$. For $k \geq 2$: $k(k-1) \geq 2 > 1$, so the inequality holds (and definitely for $k \geq 4$). Thus $(k+1)! > (k+1)^2$. ✓

Topic 2 — Run-of-Heads Recursion

Condition on the last block

- Let $p(n)$ = probability of **no run of k heads** in n fair flips.
- Look at the *tail* of the sequence. It must end in one of these blocks (otherwise a run of k forms): T, HT, HHT, ..., $\underbrace{H\cdots H}_{k-1}T$.
- Each tail of length j has probability $1/2^j$, and the rest (length $n-j$) must independently have **no run of k**.
- **Recursion:**

$$p(n) \;=\; \tfrac{1}{2}\,p(n-1) \;+\; \tfrac{1}{4}\,p(n-2) \;+\; \cdots \;+\; \tfrac{1}{2^k}\,p(n-k).$$

Initial conditions: $p(0) = p(1) = \cdots = p(k-1) = 1$ (too few flips to even form a run of k).

Worked Example — Run of 2 heads (k = 2)

Recursion: $p(n) = \frac{1}{2}\,p(n-1) + \frac{1}{4}\,p(n-2)$ for $n \geq 2$.

Initial: $p(0) = 1, p(1) = 1$.

Table:

$$p(2) = \tfrac{1}{2}(1) + \tfrac{1}{4}(1) = \tfrac{3}{4}, \qquad p(3) = \tfrac{1}{2}\cdot\tfrac{3}{4} + \tfrac{1}{4}\cdot 1 = \tfrac{5}{8}, \qquad p(4) = \tfrac{1}{2}\cdot\tfrac{5}{8} + \tfrac{1}{4}\cdot\tfrac{3}{4} = \tfrac{5}{16} + \tfrac{3}{16} = \tfrac{1}{2}.$$

Sanity check: of the 4 length-2 sequences only HH is bad, so $p(2) = 3/4$. ✓

Practice — Topic 2

2(a) Set up the recursion for "no run of 3 heads" in n flips of a fair coin.

ANSWER

Tail must end in one of: T, HT, HHT. So

$$p(n) \;=\; \tfrac{1}{2}\,p(n-1) \;+\; \tfrac{1}{4}\,p(n-2) \;+\; \tfrac{1}{8}\,p(n-3), \qquad n \geq 3.$$

2(b) Give the initial conditions $p(0), p(1), p(2)$ for the "no run of 3 heads" problem.

ANSWER

$p(0) = p(1) = p(2) = 1$. With fewer than 3 flips you can never produce a run of 3 heads.

2(c) Compute $p(5)$ for the "no run of 2 heads" problem.

ANSWER — EXTEND THE EXAMPLE'S TABLE

From the example: $p(0) = 1,\ p(1) = 1,\ p(2) = \frac{3}{4},\ p(3) = \frac{5}{8},\ p(4) = \frac{1}{2}$. Then

$$p(5) = \tfrac{1}{2}\,p(4) + \tfrac{1}{4}\,p(3) = \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{4} \cdot \tfrac{5}{8} = \tfrac{1}{4} + \tfrac{5}{32} = \tfrac{8+5}{32} = \boxed{\frac{13}{32}}.$$

Topic 3 — Series (Σ notation)

Four formulas you must own

- **Arithmetic:** $S_n = \dfrac{(a_1 + a_n)\,n}{2} = \dfrac{n}{2}\big(2a_1 + (n-1)d\big)$.
- **Geometric (finite):** $S_n = a_1 \cdot \dfrac{1 - r^n}{1 - r} \quad (r \neq 1)$.
- **Infinite geometric:** if $|r| < 1$, $S_\infty = \dfrac{a_1}{1 - r}$. If $|r| \geq 1$, the series diverges.
- **Domain of Convergence:** when $r = f(x)$, solve $|f(x)| < 1$ for x. The solution set is where the series converges; on that set the sum is $a_1/(1 - f(x))$.

Worked Example — Domain of Convergence

Find all x for which $3 + 3(x^2 - 3) + 3(x^2 - 3)^2 + \cdots$ converges, and find its sum.

Identify the ratio: $r = x^2 - 3$, $a_1 = 3$.

Solve $|r| < 1$:

$$|x^2 - 3| < 1 \iff -1 < x^2 - 3 < 1 \iff 2 < x^2 < 4 \iff \sqrt{2} < |x| < 2.$$

So $x \in (-2,\ -\sqrt{2}) \cup (\sqrt{2},\ 2)$.

Sum (on the convergence set):

$$S = \frac{3}{1 - (x^2 - 3)} = \frac{3}{4 - x^2}.$$

Practice — Topic 3

3(a) Find the sum: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$.

ANSWER — INFINITE GEOMETRIC

$a_1 = 1,\ r = \frac{1}{2},\ |r| < 1$, so

$$S = \frac{1}{1 - \frac{1}{2}} = \boxed{2}.$$

3(b) Find all x for which $5 + \frac{5(x+1)}{2} + \frac{5(x+1)^2}{4} + \cdots$ converges, and write its sum on that set.

ANSWER — DOMAIN OF CONVERGENCE

The ratio is $r = \frac{x+1}{2}$.

$$|r| < 1 \iff |x+1| < 2 \iff -3 < x < 1.$$

Sum on $(-3, 1)$:

$$S = \frac{5}{1 - \frac{x+1}{2}} = \frac{5 \cdot 2}{2 - (x+1)} = \boxed{\frac{10}{1-x}}.$$

3(c) Arithmetic: with $a_1 = 3$ and common difference $d = 4$, find S_{20}.

ANSWER — ARITHMETIC SUM

$a_{20} = a_1 + 19d = 3 + 76 = 79$. Then

$$S_{20} = \frac{(a_1 + a_{20}) \cdot 20}{2} = \frac{(3+79) \cdot 20}{2} = \boxed{820}.$$

Topic 4 — Bernoulli's Madness

Power sums · Hockey stick · Telescoping · Linear recurrence

Power sums:

- $\sum_{k=1}^{n} k^0 \;=\; n$
- $\sum_{k=1}^{n} k \;=\; \binom{n+1}{2} \;=\; \frac{n(n+1)}{2}$
- $\sum_{k=1}^{n} k^2 \;=\; \frac{n(n+1)(2n+1)}{6} \;=\; 2\binom{n+1}{3} + \binom{n+1}{2}$
- $\sum_{k=1}^{n} k^3 \;=\; \binom{n+1}{2}^2 \;=\; \left[\frac{n(n+1)}{2}\right]^2$

Hockey-stick identity (combinatorial reformulation):

$$\sum_{k=0}^{N} k(k+1)(k+2)\cdots(k+j-1) \;=\; j!\binom{N+j}{j+1}.$$

Telescoping: rewrite each term as $a_k = f(k) - f(k+1)$; the sum collapses to $f(\text{first}) - f(\text{last}+1)$.

Linear recurrence $T_n = a\,T_{n-1} + b$ with T_0 given:

$$T_n \;=\; a^n T_0 \;+\; b\cdot\frac{1-a^n}{1-a} \qquad (a \neq 1).$$

Worked Example — $\sum_{k=1}^{50} k^2$, two ways

Direct formula:

$$\sum_{k=1}^{50} k^2 = \frac{50\cdot 51\cdot 101}{6} = 42{,}925.$$

Hockey-stick check:

$$2\binom{51}{3} + \binom{51}{2} = 2\cdot\frac{51\cdot 50\cdot 49}{6} + \frac{51\cdot 50}{2} = \frac{124{,}950}{3} + 1275 = 41{,}650 + 1275 = 42{,}925. \checkmark$$

▸ Practice — Topic 4

4(a) Closed form: $\sum_{k=1}^{N} k(k+1)(k+2)$.

ANSWER — HOCKEY STICK WITH J = 3

$$\sum_{k=0}^{N} k(k+1)(k+2) \;=\; 3!\binom{N+3}{4} \;=\; 6\cdot\frac{(N+3)(N+2)(N+1)N}{24} \;=\; \boxed{\frac{N(N+1)(N+2)(N+3)}{4}}.$$

(The k = 0 term is 0, so the sum from k = 1 has the same value.)

4(b) Telescoping: closed form for $\sum_{k=1}^{N} \frac{1}{k(k+1)}$.

ANSWER — PARTIAL-FRACTION TELESCOPE

Split: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$. Then

$$\sum_{k=1}^{N}\left(\frac{1}{k} - \frac{1}{k+1}\right) = \left(1-\tfrac{1}{2}\right) + \left(\tfrac{1}{2}-\tfrac{1}{3}\right) + \cdots + \left(\tfrac{1}{N} - \tfrac{1}{N+1}\right) = 1 - \tfrac{1}{N+1} = \boxed{\frac{N}{N+1}}.$$

4(c) Solve $T_n = 2T_{n-1} + 3,\ T_0 = 1$. Find T_5 and the closed form for T_n.

ANSWER — LINEAR RECURRENCE (A = 2, B = 3)

Closed form:

$$T_n = 2^n T_0 \;+\; 3\cdot\frac{1-2^n}{1-2} = 2^n + 3(2^n - 1) = 4\cdot 2^n - 3 = \boxed{2^{n+2} - 3}.$$

Sanity: $T_0 = 4 - 3 = 1$ ✓; $T_1 = 2(1) + 3 = 5$ and $2^3 - 3 = 5$ ✓.

Then:

$$T_5 = 2^7 - 3 = 128 - 3 = \boxed{125}.$$

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